

SUMMARY

PhD candidate in privacy-preserving ML with hands-on experience building ML models, LLM systems, and production web applications. Seeking a data science internship to apply strong analytical, engineering, and problem-solving skills to real-world data challenges.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java, Bash
ML & Data Sci: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
Privacy ML: Federated Learning, Differential Privacy (DP-SGD)
Databases & Tools: MySQL, SQLite, MongoDB, Docker, Git, CI/CD

EXPERIENCE

PhD Researcher — Privacy-Preserving ML **Sep. 2025 – Present**
Eötvös Loránd University (ELTE) *Budapest, Hungary*

- Designing federated learning pipelines with differential privacy (DP-SGD) for synthetic data generation, achieving $\epsilon = 1$ privacy guarantees while preserving model utility
- Building reusable evaluation pipelines to benchmark privacy-utility tradeoffs across models and datasets, enabling reproducible research
- Developing and delivering a lab on synthetic data generation with differential privacy for the Data Security course; mentoring 2 graduate students on DP-SGD and adversarial ML
- Awarded the Stipendium Hungaricum Scholarship — a competitive Hungarian government grant for academic excellence and research potential

Software Engineer **Dec. 2024 – Mar. 2025**
RayaneBio *Remote*

- Architected and deployed a responsive e-commerce platform using React and Next.js, serving organic product customers
- Implemented secure user authentication and database models, improving user data management and transaction security
- Designed intuitive navigation and showcased customer testimonials, increasing client engagement and brand trust

PROJECTS

MyVellum **Dec. 2025 – Present**
LLM Orchestration, RAG, Docker, CI/CD

- Architected an LLM orchestration system with modular prompt templates for an AI-powered autobiography platform
- Developed vector-backed RAG pipelines with context-stitching, reducing hallucination rates in generated responses
- Integrated automated evaluation gates into CI/CD for relevance and safety metrics, ensuring quality across deployments
- Owned end-to-end API design and Docker deployment; mentored two engineers on prompt engineering best practices

WAU-Net **Jan. 2025 – Jun. 2025**
Master's Thesis — Python, PyTorch, OpenCV

- Designed a custom deep learning architecture for hair segmentation, achieving a 95.37% Dice coefficient on the benchmark dataset
- Engineered a data augmentation pipeline that improved generalization across diverse image conditions, reducing overfitting by 15%
- Evaluated model performance with precision, recall, and F1-score; optimized inference pipeline for real-time deployment

EDUCATION

PhD in Informatics — Privacy-Preserving Machine Learning **Sep. 2025 – Expected 2029**
Eötvös Loránd University (ELTE) *Budapest, Hungary*
M.Sc. in Artificial Intelligence (GPA: 3.7/4.0) **Sep. 2023 – Jun. 2025**
University of Khenchela *Khenchela, Algeria*
B.Sc. in Mathematics and Computer Science (GPA: 3.9/4.0) **Sep. 2020 – Jun. 2023**
University of Khenchela *Khenchela, Algeria*

CERTIFICATIONS & TRAINING

- **Software Engineering Specialization** (12-Month Program, Back-end Focus) — ALX Africa Feb. 2025
- **AI Career Essentials (AiCE)** — ALX South Africa Jul. 2024

LANGUAGES

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)